Mohamed Rauf Abdur Rahman Musharraf

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Portfolio: armusharraf.me | GitHub: musharraf-mac | LinkedIn

Profile

Information Technology undergraduate with hands-on experience in full-stack development, data preprocessing, and AI research projects. Builds REST APIs and web applications using Spring Boot, Next.js, and database technologies, with Python and NumPy experience for data cleaning. Has led group development projects and delivered a tourism website for a freelance client.

Education

BSc (Hons) in Information Technology 2024 – Present
Horizon Campus *Malabe, Sri Lanka*
Current GPA: 3.256/4.00 | Five semesters completed

Diploma in Software Engineering 2023 – 2025
Wayamba University of Sri Lanka *Kuliyapitiya, Sri Lanka*
Average: 67/100

GCE Advanced Level – Physical Science 2018 – 2021
BT/BC/Al-Hira Maha Vidyalaya *Kattankudy, Sri Lanka*

Technical Skills

Languages: Python, Java, C#, C, JavaScript, TypeScript, PHP, Kotlin
Front-End: React, Next.js, HTML, CSS, Tailwind CSS, Three.js
Back-End: Spring Boot, Laravel, Express.js, .NET, REST APIs, JWT
Databases & Caching: MongoDB, MySQL, PostgreSQL, Firestore, Redis
Machine Learning & AI: PyTorch, scikit-learn, LangChain, Agentic AI, NLP, NumPy, pandas
DevOps: Git, GitHub, Docker, Kubernetes, CI/CD
Cloud & Deployment: GCP (Cloud Run), Azure, Netlify, Vercel, Render, InfinityFree, Supabase, Cloudinary, MongoDB Atlas
Mobile & Development Tools: Android Studio, XML, Visual Studio, Jupyter Notebook, Figma

Experience

Freelancer 2022 – Present
Self-employed *Sri Lanka*

- Developed the Lankan Tourism website for a client, presenting agency services, customer reviews, and WhatsApp contact options.
- Prepare and format books, examination papers, and other documents using Microsoft applications.

Projects

Multimodal Explainable AI for Tuberculosis Diagnosis 2026
AI research tool | Two-person project

- Co-developed a research tool for tuberculosis diagnosis using a DenseNet convolutional neural network (CNN), with Grad-CAM to explain model predictions.
- Built an NLP model to process clinical inputs such as cough symptoms, using SHAP to explain its outputs.

Technologies: Python, PyTorch, deep learning, DenseNet, pandas, Jupyter Notebook, NLP, SHAP, Grad-CAM
GitHub: Detective-XAI-For-TB | Live demo: detective-xai-for-tb.streamlit.app

Lankan Tourism Website 2026
Client website

- Built a tourism agency website showcasing services and allowing customers to view and submit reviews.
- Connected service links and contact buttons to the agency's official WhatsApp number.

Technologies: Next.js, TypeScript, Tailwind CSS, Firestore, GCP
Website: lankantourism.lk

Projects (continued)

Health System 2026

Microservices web application | Group project | Team Lead & Full-Stack Developer

- Led the team and contributed to full-stack development of a health platform for doctor bookings, patient treatment updates, and patient profiles.
- The platform includes a public BMI calculator, BMI history for logged-in users, and AI-generated summaries of laboratory reports.
- Deployed the front end on Netlify, back-end services on GCP Cloud Run, and the database on MongoDB Atlas.

Technologies: Next.js, TypeScript, Tailwind CSS, Spring Boot, MongoDB, JWT, Redis, Docker
GitHub: SOC-AI-Health-Web-System | Live demo: health-web-system.netlify.app

Pulse Health Tracker 2026

Android application | Group project | Team Lead, Designer & Developer

- Led, designed, and contributed to development of an Android app for tracking health conditions and setting daily medication reminders.

Technologies: Kotlin, XML, Android Studio, Figma
GitHub: MyHorizonTeam/Pulse-HealthTracker

Personal Portfolio 2026

Website

- Built a personal website presenting skills, education, and projects, with an AI chatbot.

Technologies: React, JavaScript, Three.js, Tailwind CSS, AI chatbot
Website: armusharraf.me

Movie Ticket Booking Application 2024

Desktop application

- Developed a desktop application for booking movie tickets using object-oriented programming (OOP) principles.

Technologies: C#, .NET, Visual Studio
GitHub: Cinema_booking_System_OOP

Additional projects are available on my portfolio and GitHub.

Certifications & Learning

- **Python for Beginners** – University of Moratuwa
- **AI/ML Engineer** – SLIIT
- **Intro to Programming** – Kaggle
- **TryHackMe profile:** tryhackme.com/p/Musharrafcm97

Leadership & Activities

- Executive board member of the Student Council and Sports Council.
- Member of the Leo Club, Chess Club, E-sports Club, and Cyber Security Community at Horizon Campus.
- Participated in campus chess tournaments and a marathon organized by the Management Faculty at Horizon Campus.

Languages

Tamil (Native) | English (Intermediate) | Sinhala (Very basic)